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# Nanostructured fluids for polymeric coatings removal: Surfactants affect the polymer glass transition temperature

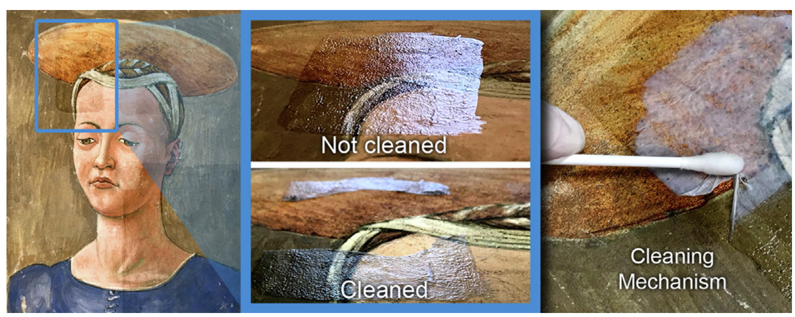
Michele Baglioni [a,1], Felipe Hidetomo Sekine [b], Taku Ogura [b,c,d], Sow-Hsin Chen [e,2], Piero Baglioni [a,e,1,*]

[a] Department of Chemistry and CSGI, University of Florence, via della Lastruccia, 3, Sesto Fiorentino (FI) 50019, Italy
[b] NIKKOL GROUP Nikko Chemicals Co., Ltd, 1-4-8, Nihonbashi-Bakurocho, Chuo-ku, Tokyo 103-0002, Japan
[c] NIKKOL GROUP Cosmos Technical Center Co., Ltd, 3-24-3 Hasune, Itabashi-ku, Tokyo 174-0046, Japan
[d] Research Institute for Science & Technology, Tokyo University of Science, 2641, Noda-shi, Chiba, Yamazaki 278-8510, Japan
[e] Department of Nuclear Science & Engineering, Massachusetts Institute of Technology, 77 Massachusetts Avenue, 24-107, Cambridge, MA 02139, USA

## HIGHLIGHTS

- Nanostructured fluids are used to remove polymer coatings in art conservation.
- Polymer films are either dewetted or swollen depending on their composition.
- Surfactants have an important role in swelling polymer films.
- Polymer swelling depends on the polymer glass temperature change induced by surfactants interaction with polymer films.
- Nonionic are more efficient than zwitterionic/ionic surfactants in lowering the $T_g$.

## GRAPHICAL ABSTRACT

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## ABSTRACT

*Hypothesis:* Nanostructured fluids (NSFs) based on water, organic solvents and surfactants are a valid alternative to the use of neat unconfined organic solvents for polymer coatings removal in art conservation. The physico-chemical processes underpinning their cleaning effectiveness in terms of swelling/dewetting of polymer films were identified as key in this context. The role of surfactants on polymers' dewetting was considered to be mainly restricted to the lowering of interfacial tensions. However, recent experiments evidenced that surfactants have an important role in swelling polymer films.
*Experiments:* Five different amphiphiles were selected, namely: sodium dodecylsulfate, dimethyldodecyl amine oxide, hexaoxyethylene decyl ether ($C_{9-11}E_6$), pentadecaoxyethylene dodecyl ether ($C_{12}E_{15}$), and methyoxypentadecaoxyethylene dodecanoate ($C_{11}COE_{15}CH_3$). They were combined with a carefully selected organic solvents' mixture (1-butanol/butanone/dimethyl carbonate) to formulate new NSFs, differing for the surfactant only, and used to perform cleaning tests on surfaces coated with Paraloid B72® and Primal AC33®. Here for the first time, polymer swelling induced by surfactants was quantified and correlated with the glass transition temperature of the two polymers by differential scanning calorimetry, before and after the exposure to the fluids. Confocal laser scanning microscopy and small-angle X-ray scattering provided additional insights on the interaction mechanism.
*Findings:* Nonionics were proven more efficient than zwitterionic/ionic amphiphiles in the polymer swelling, and, overall, methyoxy pentadecaoxyethylene dodecanoate resulted the most effective among the

* Corresponding author.
*E-mail address:* piero.baglioni@unifi.it (P. Baglioni).
[1] No kinship exists among these authors.
[2] This manuscript is dedicated to Sow-Hsin Chen, who recently passed away, with whom Piero Baglioni shared Science and a long friendship.

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selected surfactants. A direct relation between the effect of surfactants on the polymers' glass transition temperature and cleaning capacity was established. This finding, fundamental to understand the interaction mechanism between NSFs and polymer coatings or paint layers, is key to achieve a selective, effective and complete removal of polymer coatings, as recently shown in the removal of vandalism and over-paintings from street art.



## 1. Introduction

Nanostructured fluids (NSFs) are a valid alternative to traditional methods for the cleaning of art [1–4]. Surfactant-based NSFs, such as microemulsions and micellar solutions, have been applied for almost 30 years to art conservation, and they have proven particularly effective in the removal of detrimental polymeric coatings from porous inorganic substrates, as stones or wall paintings, from the majority of oil- or polymer-based paintings, or from cellulosic artifacts [1–3,5,6]. Oil-in-water (O/W) NSFs are mainly composed of water (60–90% w/w), one or more surfactants, and usually a mixture of organic solvents. Both toxicity and environmental impact of these cleaning systems are significantly lower than neat unconfined organic solvents. In fact, the amount of organic solvents included in NSFs formulations is usually reduced to 5–35%, and they are accurately selected in the class of non-toxic or limited toxicity solvents. It is worth mentioning that the cleaning mechanism of NSFs is different from that of organic solvents, i.e. the polymeric coating to be removed is swollen, softened and removed [7], sometimes involving dewetting phenomena [3,7–10], rather than being dissolved as in the case of organic solvents. The removal of the softened/dewetted coating is easily performed via a careful and gentle mechanical action, simply peeling off the NSF supporting material (e.g., a cellulose pulp poultice), or using a cotton swab, a rubber eraser or metal scalpel. This avoids the spreading and redeposition of the unwanted material into the porosity and surface of the substrate, yielding a safer and more effective cleaning action, as reported in a recent study about a novel approach for selective removal of vandalism and over-paintings from street art [11]. The present study clarifies the interaction mechanism between NSFs and polymer coatings or paint layers, which is key to achieve a selective, effective and complete removal.

It was shown that both organic solvents and surfactants play key roles in the polymeric coatings removal mechanism observed using NSFs [3,8–10]. The selection of the suitable formulations for polymer coatings removal consists of two steps: i) selection of the right organic solvents; ii) selection of surfactant. Organic solvents are important in polymers' swelling and in the lowering of their glass transition temperature, $T_g$ [9,12–14]. This, in turn, increases the mobility of macromolecular chains, and represents the first prerequisite for polymer removal. The surfactant, on the other hand, plays a crucial role in the kinetics of the process, lowering the energy cost related to the formation of new interfaces, during the first steps of dewetting [9].

There are several ways to choose the most suited organic solvents for a given polymer: tabulated solubility parameters can be used, such as Hansen [15] or Teas parameters [16]; alternatively, laboratory solubility tests with different organic solvents can be performed. Conversely, the surfactant selection is more complex. It was shown that a key parameter is represented by the surface tension, $\gamma$, of the surfactant solution, because a lower $\gamma$ should grant a smaller energy cost to start the dewetting process [3,9]. However, it was recently shown that surfactants are also able to penetrate the polymer film [3,10], contributing to its swelling [10], possibly lowering its $T_g$, as it happens with organic solvents. Therefore, the chemical nature of surfactants, and, in turn, possible specific interactions or affinity towards a given polymer, are factors that should be considered. To the best of our knowledge, studies that quantify polymers' swelling induced by surfactants included in cleaning fluids are missing. The present paper is aimed at shedding light on this subject by studying the interaction between NSFs and two polymers, i.e., poly(ethyl methacrylate/methyl acrylate) 70:30, p(EMA/MA), commercially known as Paraloid B72®, cast from an organic solvent solution, and poly(ethyl acrylate/methyl methacrylate) 67:33, p(EA/MMA), commercially known as Primal AC33®, cast from an aqueous emulsion. It was recently found that polymeric films formed from solvents' solutions tend to dewet when exposed to O/W NSFs, while films formed from polymer latexes are preferentially swollen and softened. To highlight the role of surfactants in the polymer removal mechanism, a set of NSFs based on the same solvents and differing only for the nature of the surfactant were formulated, studying their phase behavior in the 10 – 40 °C temperature range. Five different surfactants, i.e. sodium dodecylsulfate (SDS), dimethyldodecylamine oxide (DDAO), hexaoxyethylene decyl ether ($C_{9\text{-}11}E_6$, HDE), pentadecaoxyethylene dodecyl ether ($C_{12}E_{15}$, PDE), and methyoxy pentadecaoxyethylene dodecanoate ($C_{11}COE_{15}CH_3$, MPD) were selected for the study. SDS, DDAO and HDE were selected, as they are good representatives of different surfactants' classes, i.e., ionics, zwitterionics and nonionics, respectively. In addition to these, other two nonionic surfactants were considered, PDE and MPD, which were the object of a recent work [3], where the superior cleaning properties of the methyl-capped ester nonionic MPD were shown, in comparison to its non-capped ether ethoxylate homologue, PDE.

The five selected formulations were tested in polymer removal experiments from glass slides, and the results were studied by means of gravimetric analyses, differential scanning calorimetry, DSC, to measure the $T_g$ of the polymer after the interaction with the NSFs, and small-angle X-ray scattering (SAXS), in order to investigate the nanostructure of the cleaning fluids before and after the interaction with polymeric coatings. The different kinetics of Paraloid B72® dewetting when exposed to different surfactants (PDE and MPD) was evidenced through confocal laser scanning microscopy (CLSM). Finally, Paraloid B72® and Primal AC33® coatings were removed from a laboratory fresco mockup using the most effective NSF (based on MPD), simulating a real case cleaning intervention. The obtained results are relevant to the art conservation field as well as the detergency field.

## 2. Materials and methods

### 2.1. Chemicals

Sodium dodecylsulfate (SDS, Sigma-Aldrich, assay 99%), dodecyl dimethyl amine oxide (DDAO, Sigma-Aldrich, 30% aqueous solution), $C_{9\text{-}11}E_6$, hexaoxyethylene decyl ether (HDE, Croda, assay 98%), $C_{11}(C{=}O)E_{15}\text{-}CH_3$, methoxy pentadecaoxyethylene dodecanoate (MPD, Nikko Chemicals, assay 99%), $C_{12}E_{15}$, pentadecaoxyethylene dodecyl ether (PDE, Nikko Chemicals, assay 99%), butanone (methyl ethyl ketone, MEK, Sigma-Aldrich, assay 99%), 1-butanol (BuOH, Sigma-Aldrich, assay > 99%), dimethyl carbonate (DMC, Sigma-Aldrich, assay 99%), propylene carbonate (PC, Sigma-

Aldrich, assay 99%), and ethyl acetate (EA, Sigma-Aldrich, ACS Reagents, assay $\geq$ 99.5%) were used without further purification. Rhodamine B isothiocyanate (Sigma-Aldrich, assay > 99%) was used as the fluorescent dye. Water was purified with a Millipore Milli-Q gradient system (resistivity > 18 MΩ cm). Japanese paper (9.6 g/m$^2$), poly(ethyl methacrylate/methylacrylate) (p(EMA/MA) – Paraloid B72®) pellets, and cellulose powder (Arbocel® BC200, J. Rettenmaier & Sohne, Gmbh) were purchased from Zecchi, Florence. Poly(ethyl acrylate/methyl methacrylate) (p(EA/MMA) – Primal AC33®) aqueous emulsion was purchased from Phase Restauro, Florence.

### 2.2. NSFs formulations

The pseudo-ternary phase diagrams of five different water/surfactant/solvent systems were studied, where *surfactant* indicates one of the selected amphiphiles, i.e., SDS, DDAO, HDE, PDE, or MPD; and *solvent* is a mixture of BuOH/DMC/MEK (1:1:1 w/w). The phase behavior of the systems was observed at 10 °C, 20 °C, 30 °C, and 40 °C, in order to select the five NSFs' formulations to be used during the experiments on polymer coatings. A significant portion of interest of the phase diagrams was studied by sampling at least 20 different samples. The samples were equilibrated in a thermostatic bath at the selected temperatures for a week and constantly kept under observation to determine whether the system was mono- or biphasic. Finally, for each system a formulation was selected in the single-phase region, in order to: i) be stable in the whole range of investigated temperatures; ii) maintain the same molar ratio between the components, with respect to the other selected formulations. In this way, the five selected formulations presented the same molar fraction of each compound in all the formulations, differing for the surfactant nature only. The composition of the five NSFs selected for the removal of the polymer layer is reported in Table S1.

### 2.3. Polymer removal tests

Paraloid B72® was dissolved in EA to get a 10% w/w solution, while Primal AC33® was diluted with water to get a 10% w/w aqueous emulsion; the initial dry content, $\approx$ 45% w/w, was measured by evaporating the water in oven until constant weight was reached. Some frosted glass slides (5 $\times$ 5 cm$^2$) were coated with a uniform layer of the two polymers solution by drop-casting. After the evaporation of EA and water, even and continuous polymeric films were obtained with about 4 mg/cm$^2$ polymer amount. Polymer removal tests were performed using cellulose pulp poultices soaked with the NSFs, placing a sheet of Japanese paper between the poultices and the polymer film. Japanese paper is a thin and low-density paper traditionally produced in Japan, and commonly used by conservators to avoid the direct contact of poultices with the surface to treat, in order to reduce the release of fibers from cellulose pulp onto the cleaned area. The poultices were left interacting for 1.5 h, and then carefully removed. The surface was gently rinsed with water to remove possible residues of surfactant adsorbed on top of glass or the residual polymer. After the complete drying of samples, the treated glass slides were weighed to obtain the % of removed polymer.

To simulate a cleaning intervention on a real work of art, the same cleaning procedure was used to remove Paraloid B72® and Primal AC33® coatings applied on a fresco mockup. The NSF based on MPD was used for the cleaning test, as it resulted the most effective according to cleaning tests performed on the glass slides. A cellulose pulp poultice soaked with water was finally left drying on the treated area to extract possible surfactant residues from the porous matrix of the fresco.

### 2.4. Differential scanning calorimetry (DSC)

DSC measurements were performed to determine the $T_g$ of the two polymer films after the interaction with the NSFs. To this aim, some glass slides were coated with the polymers (polymer amount $\approx$ 10 mg/cm$^2$), with the same procedure described in section 2.3, and immersed in 30 mL of the five selected NSFs for 1 h. The swollen/dewetted polymer films were let drying at room temperature until constant weight was reached. Finally, measurements were conducted using a DSC Q1000 from TA Instruments on small samples (2–5 mg) taken from the dry films according to the following procedure: equilibration at −50 °C; heating ramp from −50 to 60 °C at 10 °C/min; cooling ramp from 60 to −50 °C at 10 °C/min; heating ramp from −50 to 60 °C at 10 °C/min. The first heating/cooling ramp was used to equilibrate the sample, while the second heating ramp is the one used for the $T_g$ determination. Each sample was run at least twice, in order to check for reproducibility of the measurement.

### 2.5. Confocal laser scanning microscopy (CLSM)

Confocal Microscopy experiments were performed on a Leica TCS SP8 confocal microscope (Leica Microsystems GmbH, Wetzlar, Germany) equipped with a 63X water immersion objective. For CLSM experiments on p(EMA/MA)/NSFs interaction, polymer films of about 2 μm thickness were prepared by spin-coating about 200 μl of a 10% w/w p(EMA/MA) solution in EA on cover glasses (2000 rpm, 120 s). The polymer films were stained with rhodamine B isothiocyanate, co-dissolved with the polymer solution. Two NSFs were used for the experiment, which were composed of water 79.8%, surfactant 5%, PC 15.2%, and MPD or PDE were alternatively used as the surfactant. Rhodamine B isothiocyanate was excited with a DPSS solid-state laser at 561 nm. The emission of the dye was acquired with a PMT detector in the 571–630 nm range.

### 2.6. Small-angle X-ray scattering (SAXS)

SAXS measurements were performed on the NSFs before and after 1 h interaction of 30 mL of cleaning fluid with the same polymer-coated glass slides described in Section 2.4. The scattering analyses were conducted using a HECUS S3-MICRO SWAXS-camera, equipped with a Hecus System 3 2D-point collimator (min divergence 0.4 $\times$ 0.9 mrad$^2$) and two position sensitive detectors (PSD-50 M) consisting of 1024 channels with a width of 54 μm. The $K_\alpha$ radiation ($\lambda$ = 1.542 Å) emitted by a Cu anode from the Oxford 50 W microfocus source with customized FOX-3D single-bounce multilayer point focusing optics (Xenocs, Grenoble) was used, while the $K_\beta$ line was removed by a multilayer filter. The voltage was generated by the GeniX system (Xenocs, Grenoble). The sample-to-detector distance was 26.9 cm. The volume between the sample and the detector was kept under vacuum during the measurements to minimize the scattering from the atmosphere. The camera was calibrated in the small-angle region using silver behenate ($d$ = 58.38 Å). Scattering curves were obtained in the $q$-range between 0.035 and 0.5 Å$^{-1}$. The temperature control was set to 25 °C. NSFs liquid samples were contained in 1.5 mm thick quartz capillary tubes sealed with hot-melting glue. Scattering curves were corrected for the empty capillary contribution considering the relative transmission factors. Desmearing of the SAXS curves was not necessary thanks to the focusing system.

## 3. Results and discussion

### 3.1. NSFs formulation and structural characterization

The first stage of NSFs formulation is the choice of the solvents, which should be characterized by a good balance between having excellent solvent properties towards the two polymers selected, low toxicity, and low environmental impact. The Teas solubility diagram [16] is a simple and useful tool that can be adopted to predict the effectiveness of solvents' mixtures in dissolving a given substance. It is based on the rough assumption that all chemicals have the same Hildebrand solubility parameter [17]. Accordingly, their solubility behavior is determined by the relative contribution of the three Hansen components, $\delta_d$, $\delta_p$, and $\delta_h$ (dispersion forces, polar interactions and h-bonding interactions, respectively), which are additive components of the total Hildebrand parameter. They can be used to calculate the fractional values of the Teas parameters, $f_d$, $f_p$, and $f_h$, as follows:

$$f_i = \frac{\delta_i}{\delta_d + \delta_p + \delta_h} \cdot 100 \tag{1}$$

where $i = d$, $p$ or $h$. Therefore, the sum of the three fractional parameters is always 100 and solvents and can be represented on a triangular graph. The solubility areas on Teas graph of both Paraloid B72® and Primal AC33® are shown in Fig. 1. Instead of choosing a single good solvent for the two polymers, we selected a ternary mixture. Solvents mixtures are commonly employed by restorers during cleaning tests, such as the ones proposed by Feller [18], who developed a method based on toluene/cyclohexane/acetone mixtures. Working with solvents' mixtures it is possible to obtain more versatile, effective, and safe NSFs considering that: i) each given polymer "chooses" the optimal solvent mixture to interact with, thus, one single robust solvent mixture based on a wide-range action is more efficient and practical than of using several different specific formulations; ii) sometimes effective solvents may be poorly safe for the health and the environment; however, it is usually possible replacing these solvents with mixtures having the same effectiveness, but reduced toxicity and environmental impact.

The selected ternary mixture was composed by BuOH, MEK, and DMC (1:1:1 w/w), i.e., an alcohol, a ketone and an alkyl carbonate, and its position on the Teas Graph is reported in Fig. 1. Medium-length chain alcohols are relatively low-toxicity chemicals, having limited solving power towards acrylic polymers, while possessing some surface activity, which make them effective as "cosurfactants". Ketones, such as MEK, are usually excellent solvents for a wide range of substances, including acrylic polymers. Finally, alkyl carbonates are non-toxic and low-impact organic solvents, produced via *trans*-esterification reactions of alcohols and polyols, which also are relatively safe and eco-compatible. Over the last few decades, alkyl carbonates have been increasingly used as effective solvents in industrial processes, due to their versatile properties [19], including excellent solving effectiveness for acrylic films.

BuOH, MEK and DMC are all partly water-miscible (8%, 29%, 14%, respectively, at 20 °C), and the NSFs resulting from mixing the organic solvents with water and surfactants were expected to be composed of swollen micelles dispersed in a water/solvents mixture, rather than being classically-defined microemulsions, which are usually based on completely water-immiscible organic solvents. However, from an applicative standpoint, this is actually good, for at least two reasons: i) the monophasic stable region in the NSFs phase diagrams should be significantly larger than for "classic" microemulsions, making the cleaning systems more stable and robust; ii) the maximum amount of organic solvents that can be included in the cleaning systems is higher, increasing their effectiveness.

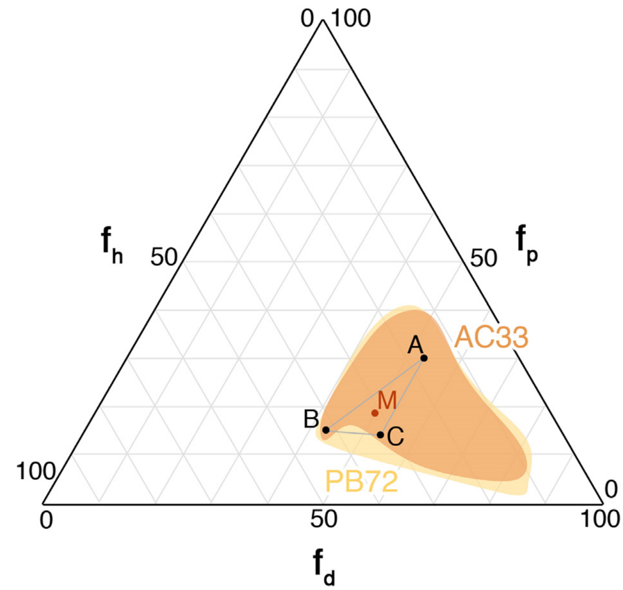


**Fig. 1.** The Teas graph reports the solubility areas of Paraloid B72® (PB72) and Primal AC33® (AC33), together with the position of the BuOH/MEK/DMC 1:1:1 (w/w) solvents' mixture (point M) selected to formulate the NSFs. Note that the position of the mixture is located well inside the polymers' solubility areas. A = MEK; B = BuOH; C = DMC.

The pseudo-ternary phase diagrams of the five different water/-surfactant/solvent systems were then studied in the 10 – 40 °C range (see **Figure S1**), where *surfactant* stands for SDS, DDAO, HDE, PDE, or MPD; and *solvent* is the BuOH/DMC/MEK (1:1:1 w/w) mixture. For each system, a formulation was selected for polymer removal experiments, which met the following criteria: i) the solvent amount in the formulation should be maximized, while the surfactant amount should be preferentially kept below 10%; ii) the formulation should be monophasic in the whole explored temperature range; iii) the molar ratio between the components in the selected formulation should be the same for each of the five systems based on the different surfactants. The composition of the five selected formulations is reported in **Table S1**.

### 3.2. Cleaning tests

The polymer removal effectiveness of the selected NSFs was evaluated as described in section 2.3. Cellulose pulp poultices soaked with the NSFs were put in contact with the polymer films for 1.5 h. In the case of Paraloid B72®, it was possible to quantify via gravimetry, the % of polymer removed by each of the five NSFs (see Fig. 2).

The film was disrupted by the action of the NSFs through a dewetting process; the polymer was swollen, almost completely detached from the glass surface, and removed together with the poultice at the end of the cleaning experiment. Note that no mechanical action or lateral friction was applied to the sample while removing the poultice from the treated surface.

Conversely, Primal AC33® films were only swollen by the five NSFs and no removal could be obtained just by lifting the poultice off the sample. Therefore, only qualitative results are available for this polymer. It is worth noting that complete Primal AC33® removal was easily achieved in almost all cases with a mechanical action, since the swollen and softened film lost its adhesion to the surface and could be peeled off from the glass. However, in conservation of precious artifacts, whenever possible, mechanical action should be avoided. Conversely, when the substrate is coherent and resistant, as in the case of a non-degraded fresco painting, this

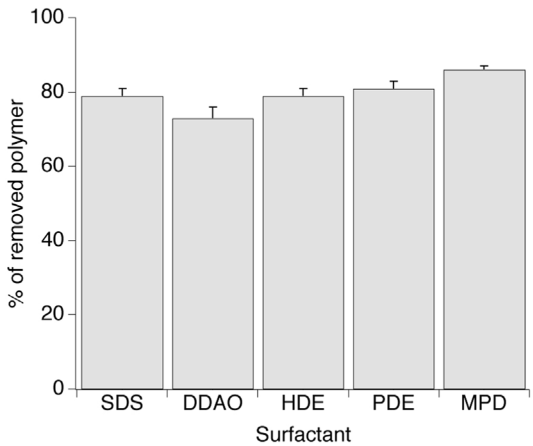


**Fig. 2.** Paraloid B72® removal tests from glass slides using the five NSFs. The % of polymer removed in a single application was measured via gravimetry. Primal AC33® data about removal tests could not be reported because this polymer is only swollen by the five NSFs and no removal could be obtained just by lifting the poultice off the sample.

cleaning mechanism is the most desirable and effective, because the complete removal of the polymer coating can be achieved very easily. Fig. 3 shows the removal of Paraloid B72® and Primal AC33® coatings from a fresco mockup using the MPD-based NSF, which resulted the most effective during cleaning tests on glass slides. Fig. 3D and 3E show the two coatings before and after the cleaning, and excellent removal was obtained for both polymers. Nonetheless, as reported for glass slides, the cleaning mechanism was different. In the case of Paraloid B72® the polymer was dewetted, disrupted, and swollen polymer droplets were removed together with the poultice. In the case of Primal AC33® the polymer was swollen, softened, and could be easily detached from the painting surface, while remaining coherent film. Fig. 3G shows the Primal AC33® coating being peeled off the surface, without leaving any residue.

As reported in the introduction section, these results are consistent with recent results on a NSF interacting with polymer films cast from solutions or from aqueous emulsions [2]. It was shown that solution-cast films are usually dewetted by the action of NSFs, while films deriving from aqueous polymer latexes are preferentially swollen and subsequently detached from the substrate, as their content in surface-active additives alter the interfacial equilibria [2]. The case of Primal AC33® is representative of this interaction mechanism, which, using a similar formalism to the term *dewetting* (i.e. the physical process opposite to wetting), could be defined *deadhering*. In this sense, NSFs can be used to *deadhere* polymer latexes from solid surfaces in the same way they can be used to *dewet* polymer films. In fact, in the absence of additives, i.e., for most of solution-cast polymer films, dewetting applies as the main mechanism responsible for NSFs cleaning effectiveness.

Thermodynamically, the tendency of a film to dewet from a substrate is described by the spreading coefficient, *S*, which accounts for the system energetic balance. In the case of the present experiments, i.e., a polymer coating deposited on a glass surface and immersed in a liquid (the NSF), *S* is expressed as follows [20]:

$$S = \gamma_{LG} - \gamma_{PG} - \gamma_{PL}$$

where $\gamma_{LG}$ is the interfacial tension between the liquid and glass, $\gamma_{PG}$ is the interfacial tension between the polymer and glass, and $\gamma_{PL}$ is the interfacial tension between the polymer and the liquid. For $S > 0$, i.e. wetting regime, films are thermodynamically stable and dewetting does not occur. On the other hand, when $S < 0$ and a fluid (or a polymer, considered as a highly viscous fluid, in this case) is only partly wetting, films are unstable or metastable and dewetting is thermodynamically favored. However, even in this case, it does not necessarily occur, as it can be hindered from a kinetics standpoint, since an energy barrier should be overcome in order to initiate the dewetting process. In the present case, where a hydrophobic polymer (Paraloid B72®) is deposited on a hydrophilic surface (glass), dewetting would be thermodynamically favored ($S < 0$), but it is kinetically inhibited due to the low mobility of macromolecular chains in the film. If chains' mobility is increased, the film becomes unstable and the polymer dewets. The main role of organic solvents in the NSF is actually to lower polymer $T_g$ below room temperature [12,14]. Once the polymer has acquired some mobility, dewetting begins with several small areas of the film randomly detaching from the substrate, with the subsequent destruction of a portion of the polymer/glass interface and the formation of new interfacial regions (polymer/liquid and glass/liquid). As a result, the total interfacial area of the system increases, and it was shown that the main role of surfactants in this process is the reduction of the energy costs related to the formation of this metastable state, by decreasing interfacial tensions [9]. However, the values of surface tension, measured at the critical micellar concentration (CMC) and reported in Table 1, for the five surfactants do not fully account for the results of Paraloid B72® removal. Fig. 2 shows that an average 80% of the polymer film was removed within 1.5 h by a single application of the NSFs. In particular all the NSFs proved to be highly effective, but, on average, nonionics seemed to be more efficient than ionic or zwitterionic surfactants, with MPD resulted as the best performing surfactant.

The high amount of organic solvents present in the NSFs tends to level up and to override the existing differences due to some possible specific effect of the surfactant in polymer removal results. Notwithstanding, these tests clearly show that *the action of the surfactant is not limited to the reduction of surface tensions.* This feature was particularly evident in the removal tests on Primal AC33®. For this polymer dewetting was not observed, and, consequently, no direct polymer removal could be achieved simply removing the poultices. However, after complete drying of the treated films, the polymer was found to be softened and swollen, with the exception of the sample exposed to the SDS-based NSF. This effect was easily attributed to the presence of residual surfactant in the Primal AC33® film, being the surfactant the only nonvolatile chemical in the NSFs' formulation.

### *3.3. Quantifying the influence of surfactants on the glass transition temperature*

It was recently found that during the NSF/polymer interaction surfactants are able to migrate into the film and work synergistically with organic solvents to induce polymer swelling [10]. The observation on the experiments here performed on Primal AC33® is in perfect agreement with these findings. It is evident that, apart from SDS, all the other surfactants are able to lower the $T_g$ of Primal AC33® well below room temperature. The effect of surfactants on the $T_g$ of polymers during the interaction with NSFs, to the best of our knowledge, was never quantified yet. To this aim, DSC measurements were performed on Paraloid B72® and Primal AC33® films immersed for 1 h in an excess of the five selected NSFs, and subsequently completely dried, in order to leave in the samples only the surfactant that migrated into the films. Gravimetric measurements were performed in order to confirm that no solvent residues were in the dried films.

The measured $T_g$ of the two pristine polymers was about 40 °C for Paraloid B72®, in agreement with the published literature [22], and about 6 °C for Primal AC33®. This value is lower than some data reported in the literature for this polymer [23], since it

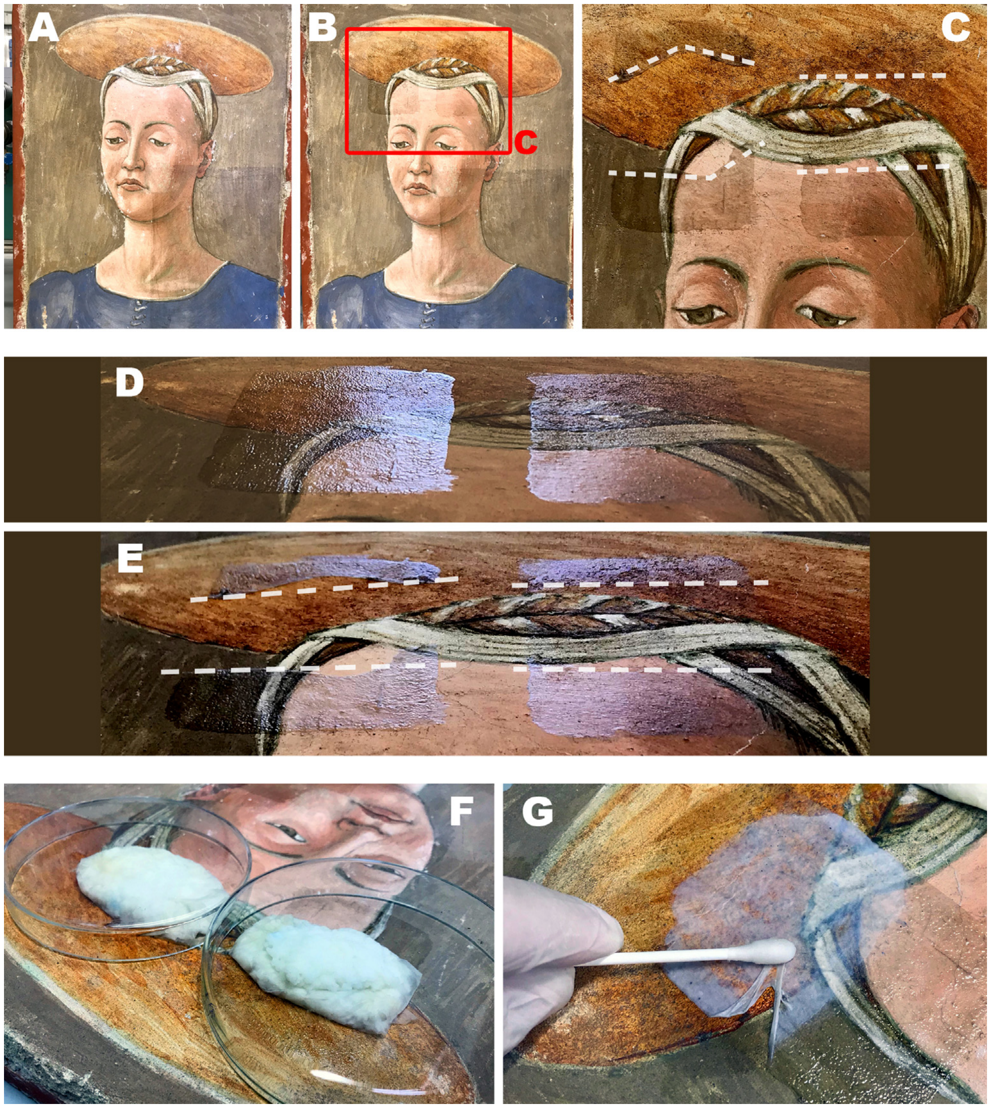


**Fig. 3.** Paraloid B72® and Primal AC33® removal tests from a laboratory fresco mockup. (A) The pristine fresco mockup. (B) The fresco mockup after the application of the two coatings (left, Primal AC33®; right, Paraloid B72®). The red box highlights the two coated areas. (C) Close up on the two coatings after the removal tests in diffuse light. The white dashed lines are a eye-guide to identify the cleaned areas. (D) Polymer coatings before cleaning, viewed in grazing light. (E) Polymer coatings after cleaning, viewed in grazing light. The white dashed lines are a eye-guide to identify the cleaned areas. (F) Cellulose pulp poultices soaked with the MPD-based NSF during the cleaning test. Petri dishes were used to cover the poultices to reduce solvents' evaporation. Two consecutive 1.5 h applications of the two poultices were performed to achieve complete polymer removal. (G) The softened (*deadhered*) Primal AC33® coating is being peeled off the fresco, leaving a completely cleaned surface.

**Table 1**
Surface tensions (measured at the CMC) for the five surfactants.

| Surfactant | CMC [mM, 25 °C] | γ (mN/m) |
|---|---|---|
| SDS | 8.3[a] | 38[a] |
| DDAO | 2.1[b] | 36[b] |
| HDE | 0.5[a] | 28[a] |
| PDE | 0.05[c] | 37.5[c] |
| MPD | 0.15[c] | 34.5[c] |

[a] From Ref. [9];
[b] at pH 5, from Ref. [21];
[c] from Ref. [3].

depends on Primal AC33® formulation. Primal AC33® is emulsified, and emulsions can be different according to the dispersing agents used. The $T_g$ values were obtained as the middle point in the baseline change observed by plotting the specific heat capacity, $c_P$, of the sample vs. temperature. The second order transition observed in the thermograms of Primal AC33® is not as clear and well defined as for Paraloid B72®. In particular, Fig. 4 shows that the onset point for the baseline change in $c_P$ for Primal AC33® is at −5.5 °C, while the offset point is at 18 °C. Therefore, in this case the first derivative of the $c_P$ ($dc_P/d_T$) could not be used to unambiguously identify the glass transition (see Fig. 4 **bottom**), which occurs over a temperature range, rather than at a specific temperature.

Fig. 5A shows the results of DSC measurements for both polymers, while **Table S2** and **Figure S2** report the complete set of DSC results. To have a second reference besides pristine polymers (samples POL in Fig. 5A), the films were exposed to the action of an aqueous mixture of BuOH, MEK and DMC (see **Table S3**) having the same composition of the average bulk phase of NSFs, according to SAXS fitting results. Interestingly (see samples SOLV in Fig. 5), the aqueous mixture had no effect on the $T_g$ of Primal AC33® dried film, while it significantly lowered the $T_g$ of Paraloid B72®. This is likely due to the fact that the $T_g$ of Primal AC33® is highly affected by the presence of a number of additives (surfactants, stabilizers,

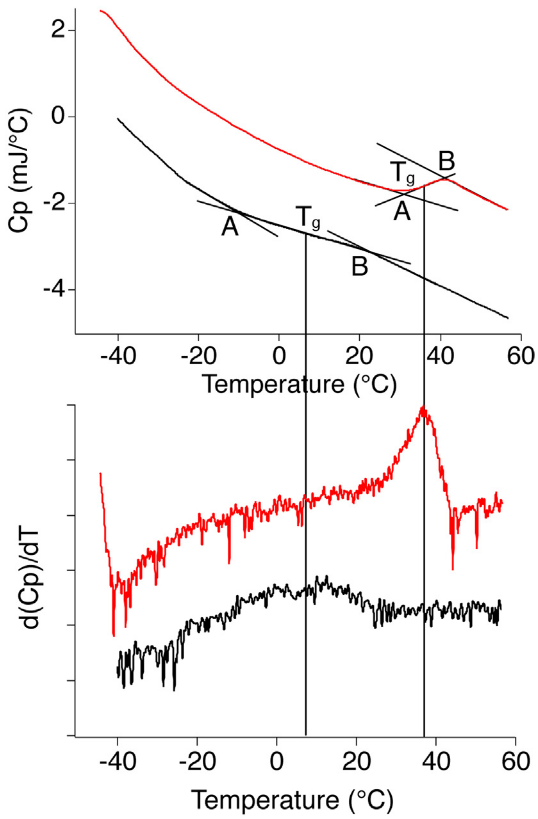


**Fig. 4.** (Top) DSC measurements performed on the sampled pristine films (red line = Paraloid B72®; black line = Primal AC33®), as detailed in section 2.4. A = onset temperature; B = offset temperature; $T_g$ = measured glass transition temperature. (Bottom) First derivative of the specific heat capacity Vs temperature; in the case of Paraloid B72®, the evident second order transition leads to a rather sharp peak in the first derivative, which can be used to unambiguously identify the $T_g$ value. Conversely, in the case of Primal AC33®, the first derivative shows only a small peak.

anti-foam agents, etc.), which already alter the original glass transition temperature of the acrylic copolymer. On the other hand, the exposure of the Paraloid B72® film to a mixture of non-solvents (water), poor solvents (BuOH), and good solvents (MEK and DMC) likely induced some polymer conformational changes that were maintained also after complete drying of the film. Overall, polymer $T_g$ reduction due to the action of surfactants is confirmed. An exception is the case of SDS-based NSF, which has no effect on the $T_g$ of both Primal AC33® and Paraloid B72®, if compared to the SOLV reference. This is in agreement with the macroscopic observation of Primal AC33® film after the cleaning experiment on glass slides. The reasons for this behavior are not straightforward, but likely are related to the ionic nature of SDS. It can be hypothesized that, once a layer of surfactant is adsorbed on the polymer surface, the negative charge present on SDS polar heads somehow inhibits its further penetration into the film.

A similar effect was observed in a previous work by Raudino et al. [24]. Four main steps are usually considered in the removal of paint or polymer-based films with cleaning formulations: i) the polymer is wetted by the cleaning fluid; ii) solvent molecules diffuse into the polymer film; iii) the film is swollen; iv) the film is removed from the solid surface. The role of surfactants is evident in the first and the last steps of this process, where lower surface tensions grant, respectively, a better wetting of the film by the cleaning fluid, and a more favorable energetic balance in dewetting/*deadhering* phenomena. However, a synergistic effect takes place between surfactants and solvents, which also affects the second and third steps of the interaction process. In fact, in the kinetic range (i.e., far from the thermodynamic equilibrium), surfactants may facilitate or hinder solvents' migration into the polymer film, as in the aforementioned case of SDS, which tend to adsorb as a charged monolayer on the polymer surface, hampering the penetration of solvents' molecules [24]. Furthermore, though solvents play a major role in this case, surfactants contribute to polymer swelling. In fact, a clear picture emerges where more or less effective surfactants can be identified. In particular, the selected surfactants follow the trend, from the most to the least effective polymer swelling capacity, i.e. MPD > HDE > PDE > DDAO > SDS. Interestingly, gravimetric measurements on the polymer films before and after the exposure to NSFs showed that the amount of surfactant adsorbed on or penetrated into the films almost perfectly correlates with the effect on the $T_g$ of the different surfactants (see Fig. 5B). The content of SDS in the dried polymer films was, on average, about 30 μg for mg of polymer, while the content of MPD was around 90 μg for mg of polymer. This clearly reflects the different influences of the two amphiphiles on the Tg of Paraloid B72® and Primal AC33®. It is clear, that the migration of nonionics into hydrophobic polymer films is favored with respect to zwitterionic/ionic amphiphiles. In fact, the most efficient surfactant in lowering the $T_g$ was MPD. MPD is a methyl-capped ester ethoxylated nonionic, which possesses excellent solubilization power towards oils and low molecular weight hydrophobic compounds [3]. It was recently shown that its enhanced hydrophobicity, due to the methyl capping of the polyoxyethylene chain and

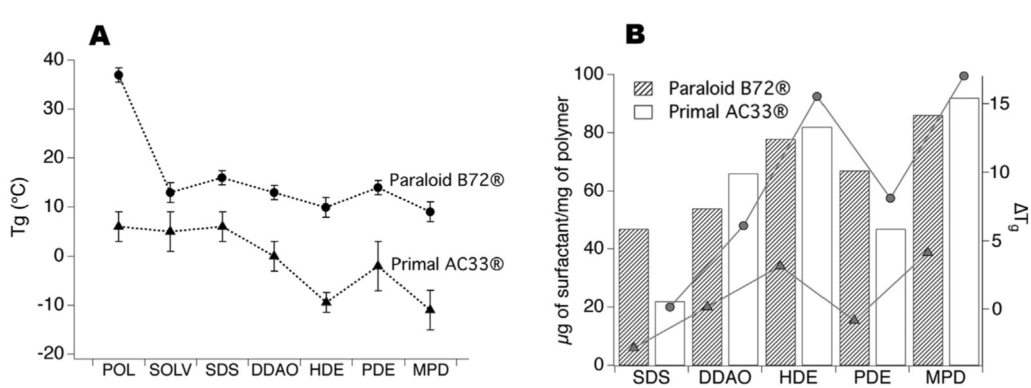


**Fig. 5.** (A) DSC measurements performed on the sampled films, as detailed in section 2.4. Samples POL are the pristine polymer films. The error in the figure was set as 10% of the ΔT between onset and offset temperatures (see **Table S2**), since the statistic uncertainty coming from the averaging of the repeated measurement was smaller. (B) The histogram shows the amount of surfactant adsorbed on and penetrated into the films exposed to the five NSFs for 1 h. Interestingly, for both polymers, the trend almost perfectly correlates with the trend of $\Delta T_g$ induced by the five different surfactants (triangles for Paraloid B72® and circles for Primal AC33®).

the peculiar conformation due to the ester bond in the middle of the molecule, makes it more efficient in interacting with hydrophobic polymers than the homologue common alcohol ethoxylate (PDE) [3]. This likely determines a favored interaction/penetration with/into the polymer film, which leads to a particularly effective lowering of polymer $T_g$. CLSM experiments highlighted this behavior (see Fig. 6). A 2 μm thick Paraloid B72® film was exposed to different ternary NSFs composed of water, propylene carbonate and, alternatively, MPD or PDE. As expected according to similar previous experiments [3,8,9], the film was completely dewetted after 20 min. However, the kinetics of the process was significantly faster for MPD. In fact, Fig. 6 shows that after 10 min of incubation with the NSFs, the film exposed to PDE was still coherent, and just presented some holes of few tenths of μm, while the one exposed to MPD already showed the thick polymer rims typical of late dewetting stages.

It is worth noting that MPD is more effective than HDE. This may be surprising, since MPD has a molecular weight that is almost two times that of HDE, thus its migration into the polymer could be less efficient (see, for instance, the comparison between HDE and PDE, which have the very same chemical structure, but different molecular size, 422 g/mol and 845 g/mol, respectively).

Since the trend in lowering the $T_g$ seems to be related to the hydrophobicity of the surfactants, their HLB (*hydrophilic to lipophilic balance*) may be an interesting parameter to consider, in order to predict their effectiveness in polymer swelling. Table 2 reports the available HLB values for the five surfactants. It was not possible to find/calculate the exact HLB values for all of them. For instance, a reliable method to calculate the HLB contribution of a methyl capping is missing; thus the HLB of MPD was considered to be significantly lower than the 15, as it is known that methyl capping induces a strong increase in the hydrophobicity of the molecule [25]. Sakamoto et al. reported on the effect of methyl capping and of the substitution of an ether with an ester bond on the cloud point of nonionic surfactant. It was found that the cloud point of the methyl ester ethoxylate was almost 20 °C lower than the one of the homologue alcohol ethoxylate [26]. This is reflected in a much lower HLB, as HLB and cloud point are related parameters [26].

Overall, the HLB values of the five surfactants account well for their efficacy in polymer swelling, i.e. amphiphiles having lower HLB are more effective in interacting with hydrophobic films. It is worth stressing the fact that if a surfactant is less efficient in polymer swelling does not mean that NSFs based on that surfactant will be ineffective as cleaning agents. Several factors work in a synergistic and complex way to confer the final effectiveness to the cleaning fluid, as it clearly appears from polymer removal tests' results, reported in Figs. 2 and 3. However, these experiments show that the surfactant nature plays an important role in determining specific interactions with polymers.

Finally, the nanostructure of the five NSFs was investigated by means of SAXS before and after the interaction with the two polymers. After testing several combinations of different form factors and structure factors, the micelles of nonionics-based NSFs were fitted as polydisperse core–shell spheres, defined by bulk/shell and shell/core contrasts, interacting through a hard spheres potential; while SDS- and DDAO-based NSFs were fitted as prolate core–shell ellipsoidal particles defined by a double contrast. In the case of SDS, a screened Coulomb potential was included in the model to account for macro-ions interparticle interaction, while DDAO micelles were considered to be non-interacting, also due to the very low surfactant concentration of this system. The scattering length densities (SLD) of bulk, shell and core, i.e., respectively, $\rho_{bulk}$, $\rho_{shell}$ and $\rho_{core}$, were calculated according to the hypothesized composition of both the micellar and the bulk aqueous phase. **Tables S4 and S5** report the fitting results for the five NSFs, while the fitting curves are shown together with experimental data in **Figure S3**.

Scattering profiles of nonionics-based NSFs could not be fitted considering the micelles as non-interacting particles, thus a structure factor, $S(q) \neq 1$, had to be included in the fitting. A hard-spheres potential was the best approximation of the intermicellar interaction for these systems. It calculates the interparticle structure factor for spherical particles interacting through excluded volume interactions, by using the Percus-Yevick [29] closure to solve the Ornstein-Zernicke equation, which describes the total correlation between two particles. In recent publications [1,11], NSFs based on higher concentrations of alcohol ethoxylates similar to HDE were fitted with a two-Yukawa [30] potential (a short range attractive potential together with a long range weak repulsive component), which is often used to reproduce realistic potentials, such as DLVO-like or Lennard–Jones-like ones [31–34], while more diluted NSFs based on PDE and MPD were fitted with a spherical form factor alone [3]. Conversely, in the present case, the two-Yukawa structure factor produced unreliable results as compared to hard-spheres potential, possibly due to the considerable amount of solvents dissolved in the aqueous bulk phase (that alters the dielectric constant of the medium) and to the SDS doping of nonionic micelles (in order to increase the systems' cloud point), which both can alter intermicellar interactions. **Figures S4-S7** report the contribution of the form and structure factors to final fitting curves, and further details on the fitting models are reported in the supporting information file (Section S3) [29,33–38].

Micelles' size and shape in all cases are in good agreement with previous SAXS and SANS analyses [1–3,7,39,40], and consistent with the molecular size of all the selected surfactants. According to fitting results, only 20 – 40 % (w/w) of the organic solvents is solubilized into the micelles, while the majority of them is dissolved in the aqueous bulk phase, as expected in view of their water-miscibility.

A clear picture emerges about the structural evolution of the cleaning fluids interacting with polymer films. The SAXS analysis confirmed that after the interaction with the polymers, the amount of organic solvents and surfactant in the NSFs was slightly depleted, and all the NSFs underwent to a minor structural reorganization, which, resulted in a small decrase in the volume fraction of scattering objects, mainly due to the surfactant migration into polymer films, usually accompanied by a minor adjustment in micelles' size. These results are consistent with macroscopic observation on the polymer films exposed to the NSFs' action and DSC analyses.

## 4. Conclusions

Five NSFs, based on five different anionic, zwitterionic and nonionic surfactants, namely: SDS, DDAO, HDE, PDE, and MPD, were investigated for polymers' removal in art conservation. The solvent mixture (BuOH/MEK/DMC 1:1:1 w/w) included in the cleaning fluids was chosen according to the solubility of the two selected polymers, i.e., Paraloid B72® (p(EMA/MA)) and Primal AC33® (p(EA/MMA)). The interaction between polymer films and NSFs was studied focusing on polymer swelling induced by the surfactant. Here, for the first time, it is shown and quantified that when a NSF interacts with a polymeric coating the synergistic action of organic solvents and surfactants is more complex than expected. In fact, *surfactants do not only affect interfacial energies, influencing both the thermodynamics and the kinetics of the cleaning process, as demonstrated by confocal microscopy investigations, but they also act as co-solvent, diffusing into the polymer film, and significantly lowering its $T_g$*, as evidenced by DSC results. This behavior is known for amphiphilic compounds, which are usually employed as plasti-

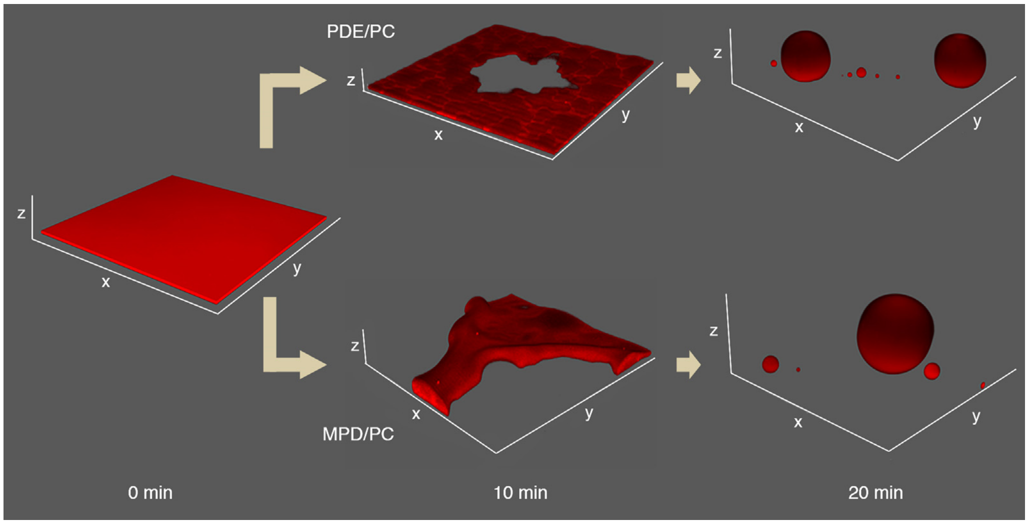


**Fig. 6.** 3D reconstructions using confocal imaging of 2 mμ thick Paraloid B72® films incubated with $H_2O$/PC/PDE (top) and $H_2O$/PC/MPD (bottom) NSFs. × and y axes are 150 μm long. The z axis has the same scale of × and y axes.

**Table 2**
HLB of the five selected surfactants.

| Surfactant | HLB |
|---|---|
| SDS | 40[a] |
| DDAO | N.A.[b] |
| HDE | 12.4[c] |
| PDE | 15.5[d] |
| MPD | <15[e] |

[a] Value taken from Kruglyakov [27];
[b] It was not possible either finding in the literature or calculating the HLB of DDAO – according to its chemical structure, it is likely comprised between 15 and 20, but it is just a guess value;
[c] Value taken from the product technical sheet;
[d] Value taken from Guo et al. [28];
[e] Value estimated according to consideration on the comparison between MPD and PDE molecular structures, according to Sakamoto et al. [26] – the actual value of MPD HLB is around 10.

cizers for plastics, and their effect on different polymers' films was widely investigated [41–45], but it was unforeseen in the context of the present study. It is still unclear whether the surfactant diffusion into the film occurs in the absence of organic solvents, which, swelling the polymer, open the way for the diffusion of the larger amphiphile molecules into the polymer. It actually seems unlikely, in agreement to previous studies showing that cleaning could not be achieved using simple surfactant/water binary as cleaning fluids [7,8]. The results reported in the present study are consistent with: i) the organic solvents easily and quickly begin to diffuse into the film; ii) when the macromolecular chains in the film are loosen, surfactants find "their way" to diffuse into the polymer, inducing a further swelling which is maintained even after solvents' evaporation, as surfactant molecules remains trapped into the film.

The behavior of different surfactants was evaluated in this paper, and it was found that uncharged (nonionic) amphiphiles are more prone to diffuse into polymer films than ionic surfactants. Moreover, surfactants' HLB can be useful to predict the polymer swelling. In particular, when hydrophobic polymers are considered, low-HLB amphiphiles are more efficient in lowering the polymer $T_g$, by diffusing into the film. MPD, methyl-capped ester nonionic surfactant, possesses excellent solubilization properties together with an increased hydrophobicity, which makes it more efficient in interacting with water-repellent polymer coatings than common alcohol ethoxylates. This work sheds light on the physico-chemical processes underlying the use of NSFs for the cleaning of works of art and will hopefully lead to an increasingly aware formulation of new classes of cleaning tools available to conservators and restorers. A recently reported example was the selective removal of vandalism and over-paintings from street art [11]. Moreover, the outcome of this study is valuable for the understanding of small molecules penetration through the skin/tissue in the wide field of cosmetic [46–50].

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## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jcis.2021.07.078.

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